HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and applied advanced **structural and thermo-mechanical FEA methodologies** for server SoCs and semiconductor packages, analyzing CPI, package assembly, warpage, and stress.
- Evaluated both **soldered and socketed packages** under various conditions including SMT, socket loading, thermal cycling, and operating conditions, identifying potential failure points.
- Analyzed **package-to-platform mechanical interaction** across SoC package, PCB, socket/retention systems, and thermal interface materials, including TIM1.5.
- Applied advanced FEA techniques such as **submodeling, contact analysis, and material nonlinearity** to complex mechanical problems.
- Developed physics-based **reliability and life-prediction models** for failure mechanisms including solder fatigue, die stress, and substrate cracking.
- Performed **sensitivity, uncertainty, and bounding analyses** to identify dominant parameters and quantify structural risk for package and platform architecture decisions.
- Partnered with package engineering teams to align **modeling assumptions, material properties, and boundary conditions** for accurate simulation outcomes.

SKILLS

FEA & Simulation – Finite Element Analysis, Structural Mechanics, Thermo-Mechanical Analysis, Semiconductor Package Mechanics, Chip-Package Interaction (CPI), Warpage Analysis, Stress Analysis, Deformation Analysis, Contact Analysis, Submodeling, Material Nonlinearity, Geometric Nonlinearity, Viscoelasticity, Creep, Fracture Mechanics, Damage Mechanics, Coupled Structural-Thermal Analysis, Reliability Modeling, Life Prediction Models, Solder Joint Reliability, Fatigue Analysis, Creep Analysis, Uncertainty Analysis, Sensitivity Analysis, Bounding Analysis, ANSYS APDL

Semiconductor Packaging – Advanced Semiconductor Packages, Soldered Packages, Socketed Packages, Die Integrity, Interconnect Integrity, BEOL Stress, Substrate Cracking, Interposer Integration, HBM Integration, 2.5D Packaging, 3D Packaging, FCBGA, FCLGA/HFCBGA, Chiplet-based Designs, Thermal Interface Materials (TIM1.5)

Platform & System Integration – Server Solutions, Data Center Applications, Qualcomm Snapdragon SoCs, Reference Platforms, Hardware/Software Solutions, PCBs, Retention Systems, Cooling Solutions, Heatsinks, Cold Plates, Platform Engineering, System Integration, Rack-Level Analyses

Programming & Scripting – Python, Shell Scripting

Data Analysis & Visualization – Power BI, Tableau, DAX, Power Query, Data Modelling

Operating Systems & Cloud – Linux, AWS (EC2, S3, Lambda, CloudWatch)

Methodologies & Tools – Agile/Scrum, Jenkins, Git, Jira, Confluence, Docker

PROJECTS

Server SoC Package Reliability Analysis

- Developed a detailed **FEA model** for a server SoC package, integrating material properties and boundary conditions from system-level requirements.
- Analyzed the impact of **thermal cycling and mechanical loading** on solder joint fatigue and die stress, quantifying expected component life.
- Generated reports detailing **stress distribution and deformation**, providing actionable feedback to package design engineers for reliability improvements.

Package-to-Platform Mechanical Interaction Study

- Modeled the mechanical interaction between a **socketed SoC package and its retention system** under thermal expansion and vibration loads.

- Investigated the influence of **thermal interface material (TIM1.5)** properties on overall system stress and warpage.
- Performed **contact analysis** to accurately simulate load transfer between the SoC package, PCB, and heatsink assembly.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)